❓ Question

You are given a CSV file **loan_approval.csv** containing loan applicant details.
Your task is to build and compare **KNN Classifier** and **Decision Tree Classifier** to predict whether the loan will be approved or not.

Perform complete machine learning workflow including data cleaning, visualization, model training, evaluation metrics, and final comparison.

📂 Dataset: `loan_approval.csv`

```
Age,Income,CreditScore,LoanAmount,Dependents,Education,SelfEmployed,PropertyA
rea,PreviousLoans,LoanApproved
25,35000,620,120000,0,Graduate,No,Urban,1,No
42,72000,780,200000,2,Graduate,No,Semiurban,0,Yes
35,58000,690,150000,1,Graduate,Yes,Rural,2,Yes
28,26000,590,100000,0,Not Graduate,No,Urban,1,No
50,85000,810,250000,3,Graduate,No,Semiurban,0,Yes
31,46000,640,130000,1,Graduate,Yes,Rural,1,No
39,67000,730,180000,2,Graduate,No,Urban,0,Yes
29,31000,610,110000,0,Not Graduate,No,Rural,2,No
45,79000,800,230000,2,Graduate,No,Semiurban,0,Yes
33,52000,675,145000,1,Graduate,Yes,Urban,1,Yes
36,,710,170000,2,Graduate,No,Urban,0,Yes
42,72000,780,200000,2,Graduate,No,Semiurban,0,Yes
27,28000,600,,0,Not Graduate,No,Rural,1,No
```

📝 Perform the Following Tasks

🔷 Part A: Load Dataset & Information (3 Marks)

1. Read CSV file using pandas.
2. Display first 5 rows.
3. Display shape of dataset.
4. Show column names and data types.

🔷 Part B: Data Cleaning (5 Marks)

1. Check missing values.
2. Fill missing values in **Income** and **LoanAmount** using median.
3. Check duplicate rows.
4. Remove duplicate rows.
5. Show cleaned dataset shape.

🔷 Part C: Preprocessing (3 Marks)

1. Convert categorical columns into numerical using `pd.get_dummies()`
2. Define:
   - `X` = independent variables
   - `y` = target column
3. Split dataset into training and testing set using 80:20 ratio.

🔷 Part D: KNN Model Training (2 Marks)

1. Train **KNN Classifier** with `n_neighbors = 3`
2. Predict test data.
3. Display:
   - Accuracy Score
   - Precision Score
   - Recall Score
   - F1 Score
   - Confusion Matrix

🔷 Part E: Decision Tree Model Training (2 Marks)

1. Train **Decision Tree Classifier**
2. Predict test data.
3. Display:
   - Accuracy Score
   - Precision Score
   - Recall Score
   - F1 Score

    - Confusion Matrix
4. Plot Decision Tree.

🔷 Part F: Final Model Comparison using DataFrame (1 Mark)

Create a pandas DataFrame:

Model	Accuracy	Precision	Recall	F1 Score
KNN	?	?	?	?
Decision Tree	?	?	?	?

Print the DataFrame and write which model performed better.

🔷 Part G: Data Visualization (4 Marks)

Create the following graphs:

1. Scatter plot of **Income vs CreditScore**
2. Boxplot of **LoanAmount**
3. Bar chart of average **Income by PropertyArea**
4. Pie chart of LoanApproved distribution